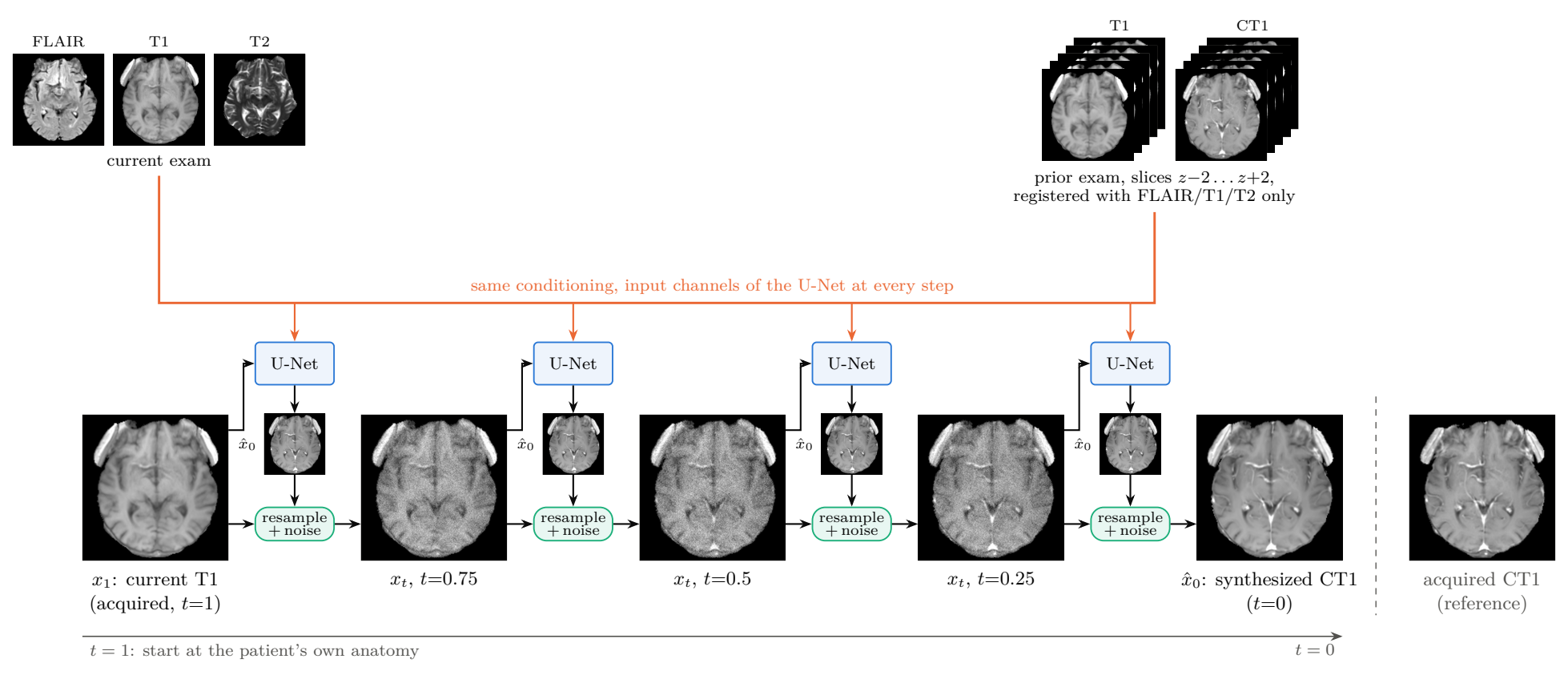

FLAIR
T1
T2
current exam
T1
CT1
prior exam, slices $z-2 \dots z+2$,
registered with FLAIR/T1/T2 only
same conditioning, input channels of the U-Net at every step
U-Net
U-Net
U-Net
U-Net
$\hat{x}_0$
$\hat{x}_0$
$\hat{x}_0$
$\hat{x}_0$
resample + noise
resample + noise
resample + noise
resample + noise
x_1: current T1
(acquired, t=1)
x_t, t=0.75
x_t, t=0.5
x_t, t=0.25
$\hat{x}_0$: synthesized CT1
(t=0)
acquired CT1
(reference)
$t = 1$: start at the patient's own anatomy
$t = 0$